

GOVERNMENT SCHOOL MANAGEMENT SYSTEM

Software Requirements Specification

SRS Document — IEEE 830 Format

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for the Government School Management System (GSMS), a centralized, web-based platform designed to digitize academic and administrative operations across government schools. This document is intended for the development team, project stakeholders, education authorities, and quality assurance personnel, and serves as the authoritative reference for what the system must do prior to design and implementation.

1.2 Scope

GSMS will provide a single, role-based platform covering Student Management, Teacher Management, Staff Management, Academic Management, Attendance, Timetable, Examination, Library, Inventory, Welfare Programs, Communication, Reporting and User Management. The system replaces manual and disconnected record-keeping with a secure, auditable, and centrally reportable digital workflow accessible to Principals, Vice Principals, School Administrators, Teachers, Administrative Staff, Librarians, Laboratory Assistants, Students, Parents and Education Officers. The initial release targets a single-school deployment model that is architecturally extensible to multi-school, district-level and Ministry-level rollouts in future phases.

1.3 Definitions, Acronyms and Abbreviations

Term	Definition
GSMS	Government School Management System
RBAC	Role-Based Access Control
JWT	JSON Web Token, used for stateless authentication
API	Application Programming Interface
REST	Representational State Transfer, an architectural style for web APIs
SRS	Software Requirements Specification
SDS	Software Design Specification
UAT	User Acceptance Testing
EO	Education Officer, a zonal/provincial education authority user
CRUD	Create, Read, Update, Delete

1.4 References

Government School Management System – Project Proposal (source document, 2026); IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications; OWASP Application Security Verification Standard (ASVS); Sri Lanka Ministry of Education examination and academic calendar guidelines.

1.5 Overview

Section 2 describes the product context, high-level functions, user classes, operating environment, constraints and assumptions. Section 3 specifies detailed functional requirements organized by module, along with external interface and non-functional requirements. Section 4 lists other requirements including data, legal and compliance considerations, and Section 5 contains appendices.

2. Overall Description

2.1 Product Perspective

GSMS is a new, self-contained web-based system that is not dependent on any existing legacy software, though it must be able to import baseline student and staff data from spreadsheet-based records during onboarding. The system follows a three-tier architecture: a React.js single-page frontend, a Node.js/Express.js REST API backend, and a PostgreSQL relational database, secured with JWT-based authentication and role-based access control. Full architectural detail is provided in the accompanying Software Design Specification (SDS).

2.2 Product Functions (Summary)

At a high level, GSMS shall allow authorized users to:

- Register, update and maintain student, teacher and staff records across academic years.
- Record and monitor daily attendance for students and staff, with automated absence alerts.
- Create, publish and manage class timetables, subject allocations and teacher workloads.
- Plan, conduct and grade examinations, and generate term/annual report cards.
- Manage library catalogues, circulation, and inventory of school assets and consumables.
- Administer welfare programs (meals, scholarships, uniforms, textbook distribution) and track eligibility and disbursement.
- Send announcements and targeted notifications to students, parents and staff.
- Generate operational, academic and compliance reports for school and education-authority use.
- Manage user accounts, roles and permissions with full audit logging.

2.3 User Classes and Characteristics

User Class	Description	Technical Proficiency
Principal / Vice Principal	Full oversight of school operations, approvals, and school-level reporting.	Low–Medium
School Administrator	Day-to-day data entry, configuration, user management, and coordination across modules.	Medium
Teacher	Attendance marking, grading, timetable viewing, communication with parents/students.	Low–Medium
Administrative Staff	Records processing, welfare administration, inventory handling.	Low–Medium
Librarian	Catalogue management and book circulation.	Low
Laboratory Assistant	Lab inventory and consumables tracking.	Low
Student	View timetable, attendance, exam results, announcements.	Low–Medium
Parent	View child's attendance, results, fee/welfare status, and receive communications.	Low
Education Officer	Read-only, aggregated cross-school reporting and compliance monitoring.	Medium

2.4 Operating Environment

GSMS shall operate as a responsive web application accessible via modern browsers (Chrome, Firefox, Edge, Safari, latest two major versions) on desktop, tablet and mobile form factors. The backend shall be deployable on Linux-based cloud or on-premises infrastructure, containerized for portability. The system shall function on standard broadband and moderate-bandwidth school network connections, with graceful degradation on slower links.

2.5 Design and Implementation Constraints

- Frontend: React.js with Tailwind CSS, as fixed by the approved technology stack.
- Backend: Node.js with Express.js, exposing RESTful APIs.
- Database: PostgreSQL as the system of record.
- Authentication: JWT-based sessions with Role-Based Access Control (RBAC).
- The system must comply with applicable Sri Lankan data protection expectations for student and staff personal data.
- The system must remain operable in low-connectivity school environments where practical (e.g. paginated data loads, offline-tolerant attendance capture as a future enhancement).

2.6 Assumptions and Dependencies

- Each school onboarding GSMS has at minimum one designated System Administrator responsible for initial data setup.
- Baseline student, teacher and staff data can be supplied in a structured spreadsheet format for bulk import.
- Schools have at least intermittent internet connectivity for cloud-hosted deployments.
- SMS/email gateway credentials will be provided by the school or Ministry for the Communication module to dispatch notifications.

3. Specific Requirements

Functional requirements are grouped by module. Each requirement is uniquely identified (FR-<Module>-<n>) to support traceability into design, test cases and UAT scripts.

3.1 User Management

Handles accounts, authentication and access control for all user classes.

ID	Requirement
FR-1-01	The system shall allow Administrators to create, edit, deactivate and delete user accounts for all user classes.
FR-1-02	The system shall enforce RBAC so that each user can only access modules and data permitted to their role.
FR-1-03	The system shall require secure login via username/email and password, issuing a signed JWT on success.
FR-1-04	The system shall support password reset via a verified email or administrator-initiated reset.
FR-1-05	The system shall log all authentication attempts and critical account changes for audit purposes.
FR-1-06	The system shall enforce configurable password complexity and session expiry policies.

3.2 Student Management

Maintains the authoritative record of every enrolled student.

ID	Requirement
FR-2-01	The system shall allow authorized users to register new students with personal, guardian and academic details.
FR-2-02	The system shall support editing, promotion (grade/year progression), transfer and de-registration of students.
FR-2-03	The system shall generate a unique student ID upon registration.
FR-2-04	The system shall allow bulk import of student records via spreadsheet upload.
FR-2-05	The system shall maintain historical academic records across multiple years for each student.
FR-2-06	The system shall allow parents to be linked to one or more student profiles.

3.3 Teacher Management

Maintains teacher profiles, subject assignments and workload data.

ID	Requirement
FR-3-01	The system shall allow registration and maintenance of teacher profiles including qualifications and subject specializations.
FR-3-02	The system shall allow assignment of teachers to classes, subjects and sections.
FR-3-03	The system shall track teacher workload (periods per week) to support timetable validation.
FR-3-04	The system shall maintain a history of teacher transfers and role changes within the school.

3.4 Staff Management

Maintains non-teaching staff records.

ID	Requirement
FR-4-01	The system shall allow registration and maintenance of administrative and support staff records.
FR-4-02	The system shall record staff roles, responsibilities and reporting lines.
FR-4-03	The system shall support staff attendance and leave tracking consistent with the Attendance module.

3.5 Academic Management

Manages grades, sections, subjects and curricula.

ID	Requirement
FR-5-01	The system shall allow configuration of academic years, grades/classes and sections.
FR-5-02	The system shall allow definition of subjects and their allocation to grades and sections.
FR-5-03	The system shall support curriculum/syllabus reference attachment per subject.

3.6 Attendance

Captures and reports daily student and staff attendance.

ID	Requirement
FR-6-01	The system shall allow teachers to mark daily/period-wise student attendance electronically.
FR-6-02	The system shall allow HR/administrative staff to mark staff attendance.
FR-6-03	The system shall automatically flag students exceeding configurable absence thresholds.
FR-6-04	The system shall notify parents automatically when a student is marked absent.
FR-6-05	The system shall generate attendance summary reports by class, section and date range.

3.7 Timetable

Builds and publishes class and teacher timetables.

ID	Requirement
FR-7-01	The system shall allow Administrators to create weekly timetables per class/section.
FR-7-02	The system shall detect and prevent teacher and room double-booking conflicts.
FR-7-03	The system shall allow publishing of finalized timetables to teachers, students and parents.
FR-7-04	The system shall support ad-hoc timetable changes (substitutions) with notification to affected users.

3.8 Examination

Manages exam scheduling, marks entry and report card generation.

ID	Requirement
FR-8-01	The system shall allow creation of examination schedules by term and subject.

ID	Requirement
FR-8-02	The system shall allow teachers to enter and edit marks within a defined submission window.
FR-8-03	The system shall automatically compute grades, averages and rankings based on configurable grading schemes.
FR-8-04	The system shall generate downloadable/printable report cards per student per term.
FR-8-05	The system shall maintain historical examination records across academic years.

3.9 Library

Manages the catalogue and circulation of library resources.

ID	Requirement
FR-9-01	The system shall allow the librarian to catalogue books and resources with unique identifiers.
FR-9-02	The system shall support issue and return transactions with due-date tracking.
FR-9-03	The system shall calculate overdue status and notify borrowers.
FR-9-04	The system shall provide search of the catalogue by title, author, subject or ID.

3.10 Inventory

Tracks school assets and consumable stock.

ID	Requirement
FR-10-01	The system shall allow recording of school assets with category, quantity, location and condition.
FR-10-02	The system shall track consumable stock levels and support low-stock alerts.
FR-10-03	The system shall record issuance of inventory items to departments or individuals.
FR-10-04	The system shall generate inventory audit reports.

3.11 Welfare

Administers welfare programs including meals, textbooks, uniforms and scholarships.

ID	Requirement
FR-11-01	The system shall allow configuration of welfare programs with eligibility criteria.
FR-11-02	The system shall allow enrolment of eligible students into applicable welfare programs.
FR-11-03	The system shall track disbursement/distribution status per student per program.
FR-11-04	The system shall generate welfare program coverage and utilization reports.

3.12 Communication

Enables announcements and targeted notifications.

ID	Requirement
FR-12-01	The system shall allow Administrators and Teachers to broadcast announcements to selected user groups.

ID	Requirement
FR-12-02	The system shall support targeted notifications (e.g. to a specific class or parent group) via in-app, email and/or SMS channels where configured.
FR-12-03	The system shall maintain a message history accessible to the sender and recipients.

3.13 Reports

Provides cross-module reporting for operational and compliance needs.

ID	Requirement
FR-13-01	The system shall provide pre-built reports for attendance, academic performance, welfare coverage and inventory status.
FR-13-02	The system shall allow filtering of reports by academic year, grade, section and date range.
FR-13-03	The system shall support export of reports to PDF and Excel formats.
FR-13-04	The system shall provide Education Officers with read-only, aggregated cross-school reporting where applicable.

3.14 External Interface Requirements

User Interfaces

GSMS shall provide a responsive web interface with role-specific dashboards. Navigation shall be consistent across modules, with a persistent sidebar/menu structure and a global search where applicable. The interface shall follow WCAG 2.1 AA accessibility guidance for color contrast and keyboard navigation.

Hardware Interfaces

No specialized hardware is required for the initial release. Future enhancements may require interfaces to QR/barcode scanners and biometric or facial-recognition attendance devices.

Software Interfaces

The system shall expose RESTful JSON APIs consumed by the React frontend. It shall integrate with an email delivery service and, optionally, an SMS gateway for the Communication and Welfare modules. It shall support data export interfaces to common office formats (CSV, XLSX, PDF).

Communications Interfaces

All client-server communication shall occur over HTTPS (TLS 1.2 or higher).

3.15 Non-Functional Requirements

Category	Requirement
Security	All data in transit shall be encrypted via TLS; passwords shall be hashed using a strong adaptive algorithm (e.g. bcrypt/argon2); access shall be enforced via RBAC at the API layer for every request.
Performance	95% of standard API requests shall complete within 2 seconds under expected concurrent load for a single-school deployment (up to ~500 concurrent users).
Scalability	The architecture shall support horizontal scaling of the API layer and shall be extensible to multi-school/district deployments without redesign.
Reliability	The system shall target 99.5% uptime during school operating hours, with automated daily database backups.
Usability	Core workflows (attendance marking, marks entry) shall be completable by a first-time user of average digital literacy within 3 guided attempts.
Maintainability	The codebase shall follow modular, documented conventions enabling a new developer to onboard to a single module within 2 working days.
Auditability	All create/update/delete operations on sensitive records (students, marks, welfare disbursement) shall be logged with user, timestamp and change detail.
Backup & Recovery	Automated daily backups with a defined Recovery Point Objective (RPO) of 24 hours and Recovery Time Objective (RTO) of 8 hours.
Compatibility	The system shall render correctly on the latest two major versions of Chrome, Firefox, Edge and Safari, and on common tablet screen sizes.

4. Other Requirements

4.1 Data Requirements

Student, teacher and staff personal data shall be classified as sensitive and access-restricted accordingly. Examination marks and welfare eligibility data shall be retained for a minimum of the applicable academic record-retention period defined by the school/Ministry. Data import/export shall preserve referential integrity between students, guardians, classes and academic years.

4.2 Legal, Regulatory and Compliance Requirements

The system shall be designed to support compliance with applicable Sri Lankan data protection requirements for the handling of minors' personal data, and shall support future integration with Ministry of Education reporting formats as they are defined.

4.3 Traceability

Each functional requirement ID (FR-<Module>-<n>) shall be traceable to a corresponding design element in the Software Design Specification and to at least one test case in the QA test plan, as described in the accompanying Development Methodology document.

4.4 Future Enhancements (Out of Current Scope)

- Native mobile applications for parents, students and teachers.
- QR-code based student ID cards for attendance and identification.
- AI-driven analytics for academic performance and dropout-risk prediction.
- Automatic, constraint-based timetable generation.
- Facial-recognition based attendance capture.
- Direct integration with Ministry of Education central systems.

5. Appendix — Requirement Priority Key

Priority	Meaning
Must Have	Required for initial release (MVP); system is not viable without it.
Should Have	Important but not launch-blocking; targeted for early follow-up release.
Could Have	Desirable enhancement, scheduled opportunistically.

Note: All functional requirements listed in Section 3 are classified as Must Have for the initial release unless otherwise noted during sprint planning, as detailed in the Development Methodology document (GSMS-MET-001).